California Traffic Collision Analysis Report

This report summarizes collision patterns in the California traffic collision project. It looks at severity, people involved, weather, lighting, weekdays, locations, and time.

Main findings

Los Angeles County had the largest number of recorded collisions. Property-damage-only records made up the largest severity group. Clear weather and daylight were the most common recorded conditions.

Two saved results need checking before use: the fatality query returned 0.0%, and the hourly query placed every record at midnight. The hourly code appears to use a date field without the separate collision time.

Key results

Collision records: 780,981

Top county: Los Angeles - 253,556

Top month: October - 69,599

Top weather: Clear - 647,360

Top lighting: Daylight - 523,519

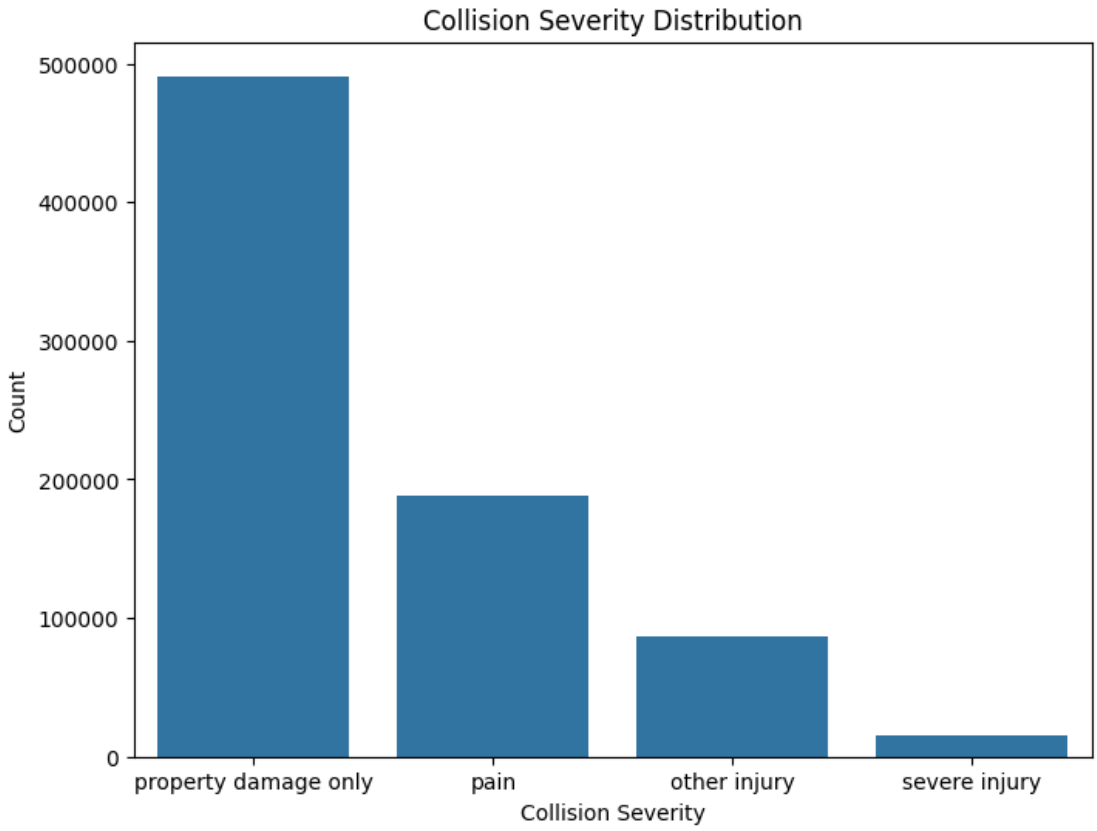
Most common hour: 00:00

1. **Data and preparation**

The notebook describes the source as California collision data from SWITRS. It loads collision, party, victim, and case ID tables and uses PySpark to prepare them.

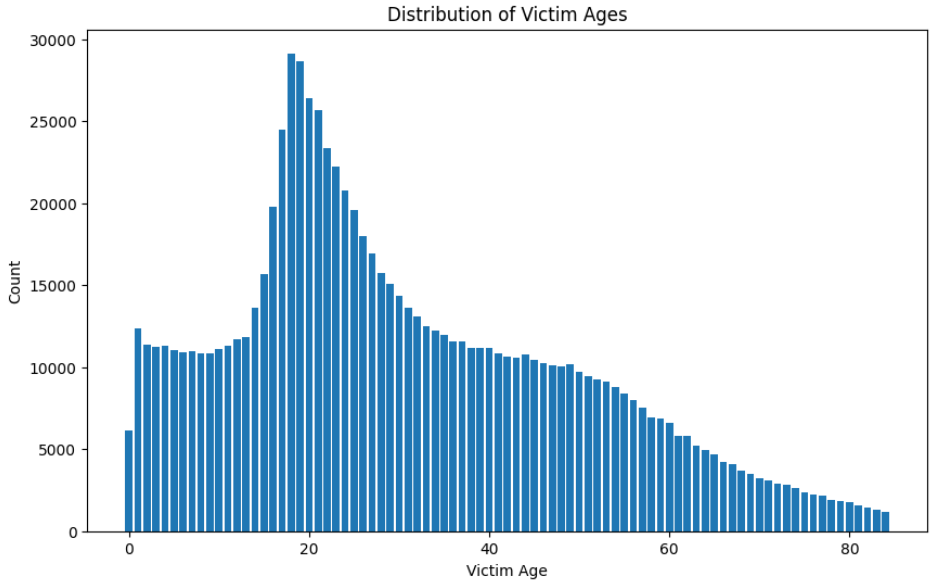
- Rows missing IDs needed for joins were removed.
- Missing text values were filled with “Unknown.” Other missing values were left as missing.
- Duplicate rows were removed using case or record IDs.
- No columns were dropped by the 80% missing-value rule.
- Numerical outliers were checked with the interquartile range rule. Columns with a zero range were left unchanged.
- Some location fields had many missing values, so the location plots include only records with valid coordinates.

1. **Distribution of Collision severity**
   Property-damage-only collisions were the largest group. Pain-related injuries were next, followed by other injury and severe injury records.

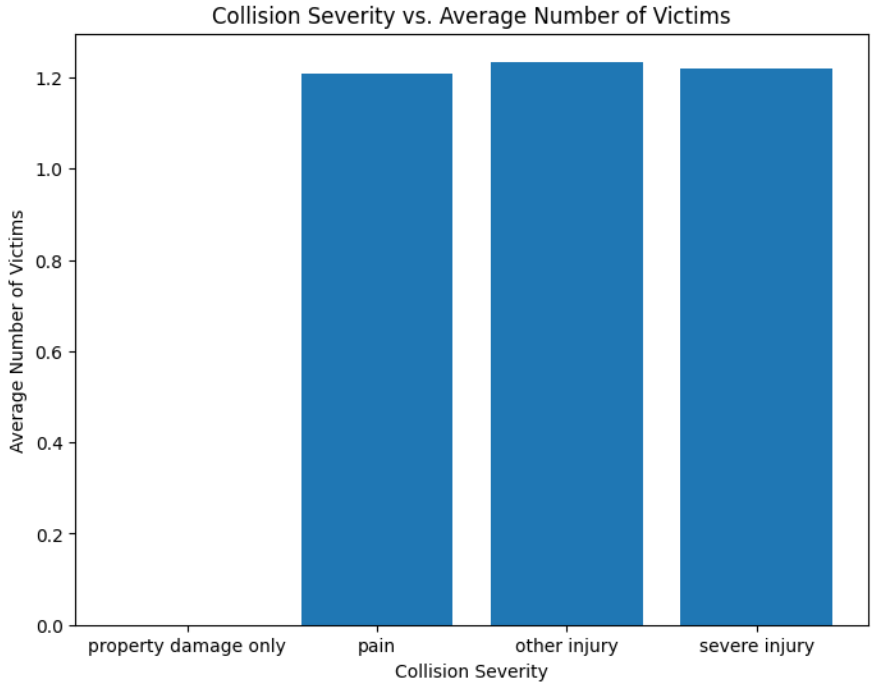

Collision Severity Distribution
500000
400000
300000
200000
100000
0
Count
property damage only
pain
other injury
severe injury
Collision Severity

2. **Victim age and severity**

The victim-age chart rises sharply in the late teens, peaks around ages 18 to 20, then generally declines with age.

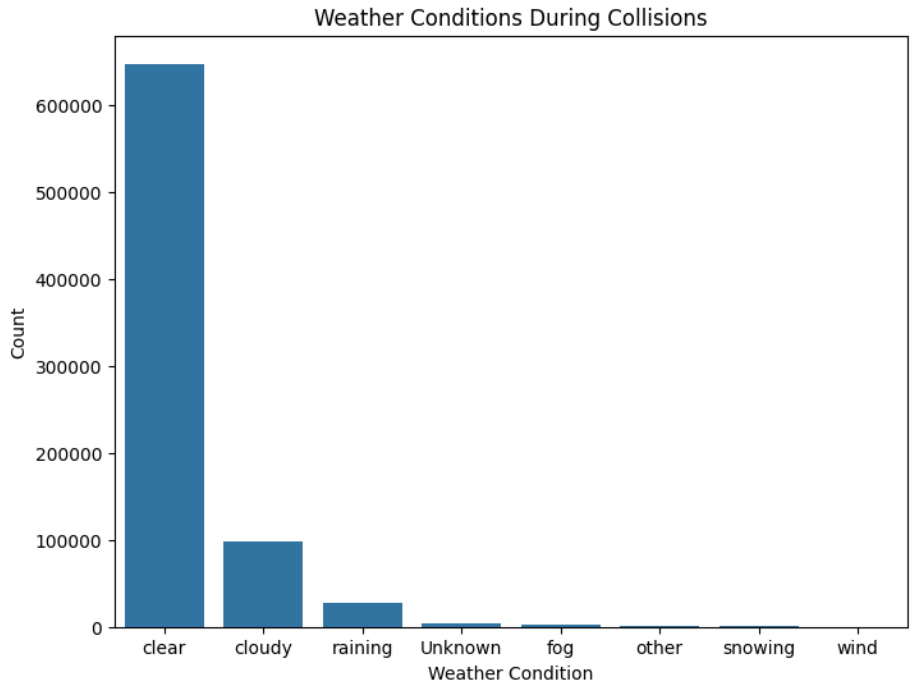


For injury categories, the average number of killed and injured victims is close to 1.2. The property-damage-only group has no visible bar in this chart.



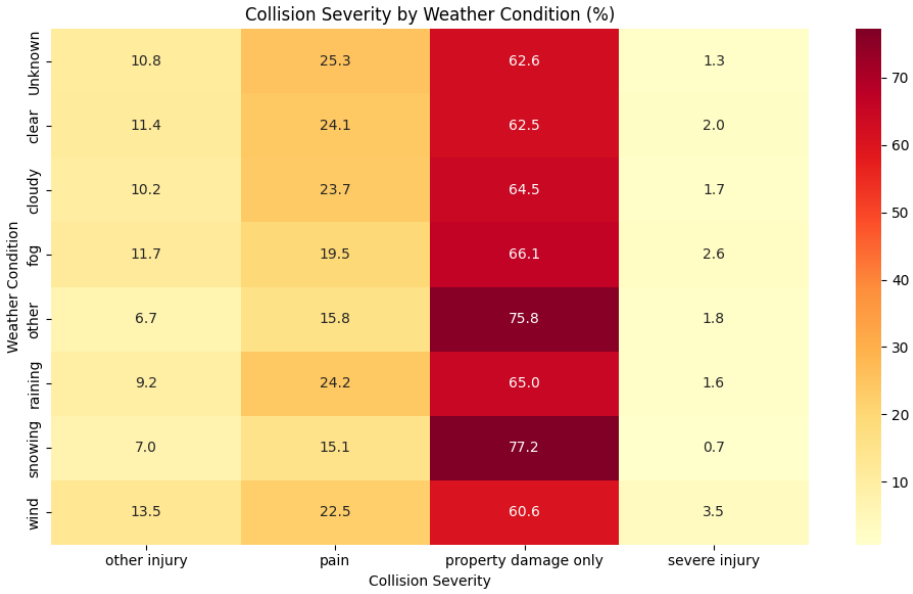
3. Weather and collision severity

Clear was the most common weather condition, with 647,360 collisions. Cloudy weather was second.



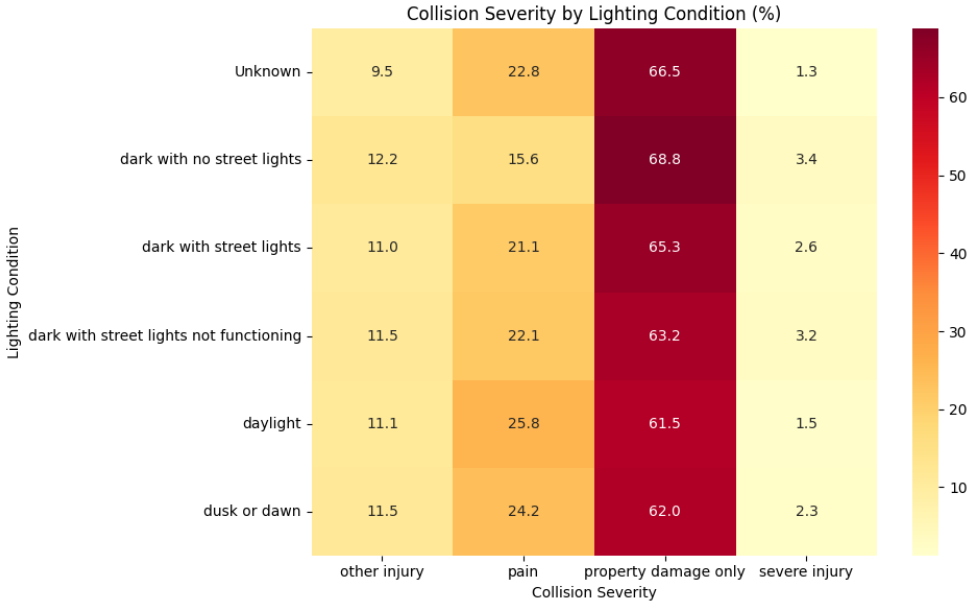
Collision counts by recorded weather condition.

The heatmap shows that property-damage-only records were the largest share in every weather category. Each row shows the severity mix within that weather category.

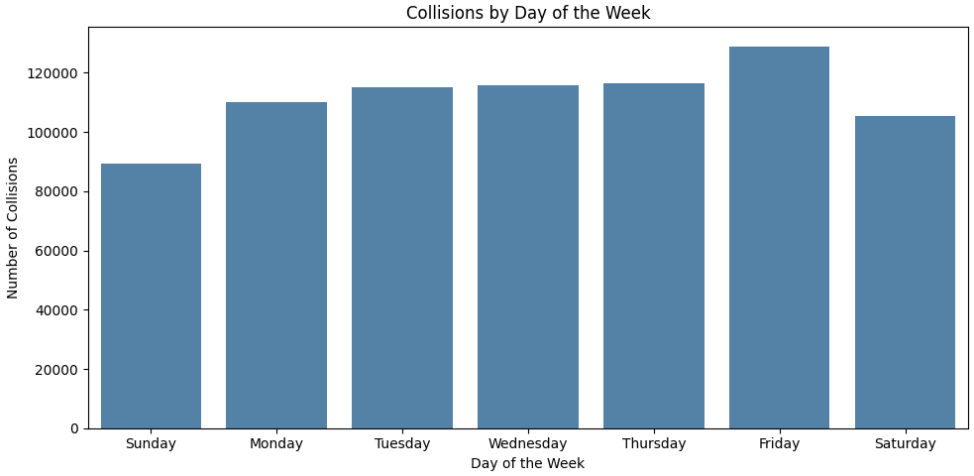


4. **Lighting and weekday patterns**

Daylight had the most collisions by count (523,519). Dark conditions with street lights had 177,306, and dark conditions without street lights had 46,904. Property-damage-only collisions were the largest share across lighting groups.



Friday had the highest weekday count in the chart. Sunday had the lowest. Monday through Thursday were at similar levels. Traffic volume may differ between weekdays.



5. County patterns

Los Angeles County recorded 253,556 collisions, far more than any other county. The next four were Orange, San Bernardino, San Diego, and Alameda.

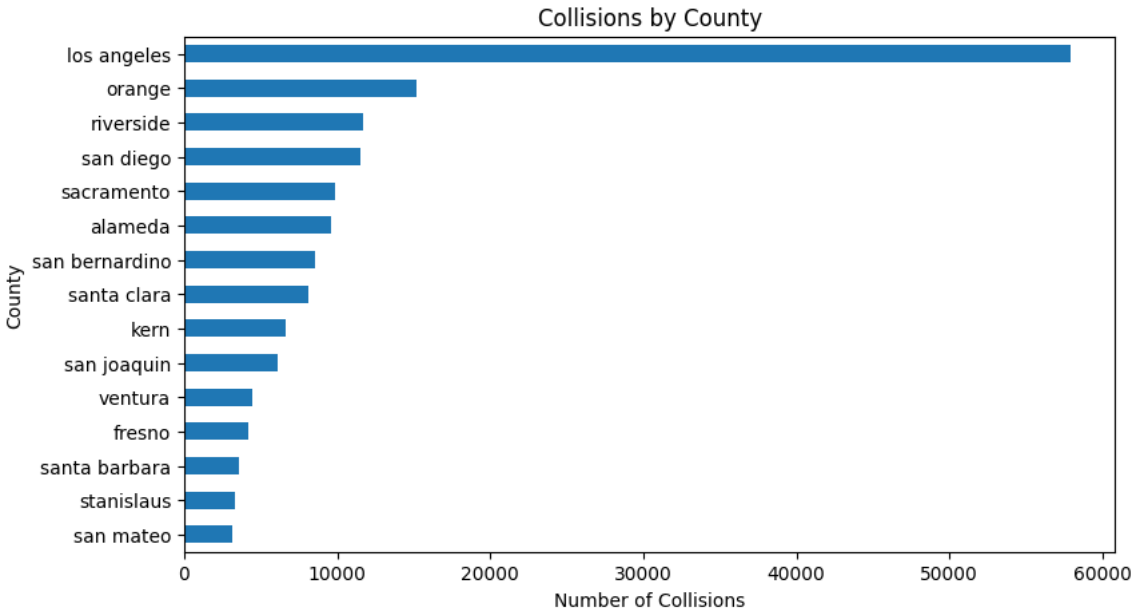
Los Angeles: 253,556

Orange: 62,734

San Bernardino: 44,830

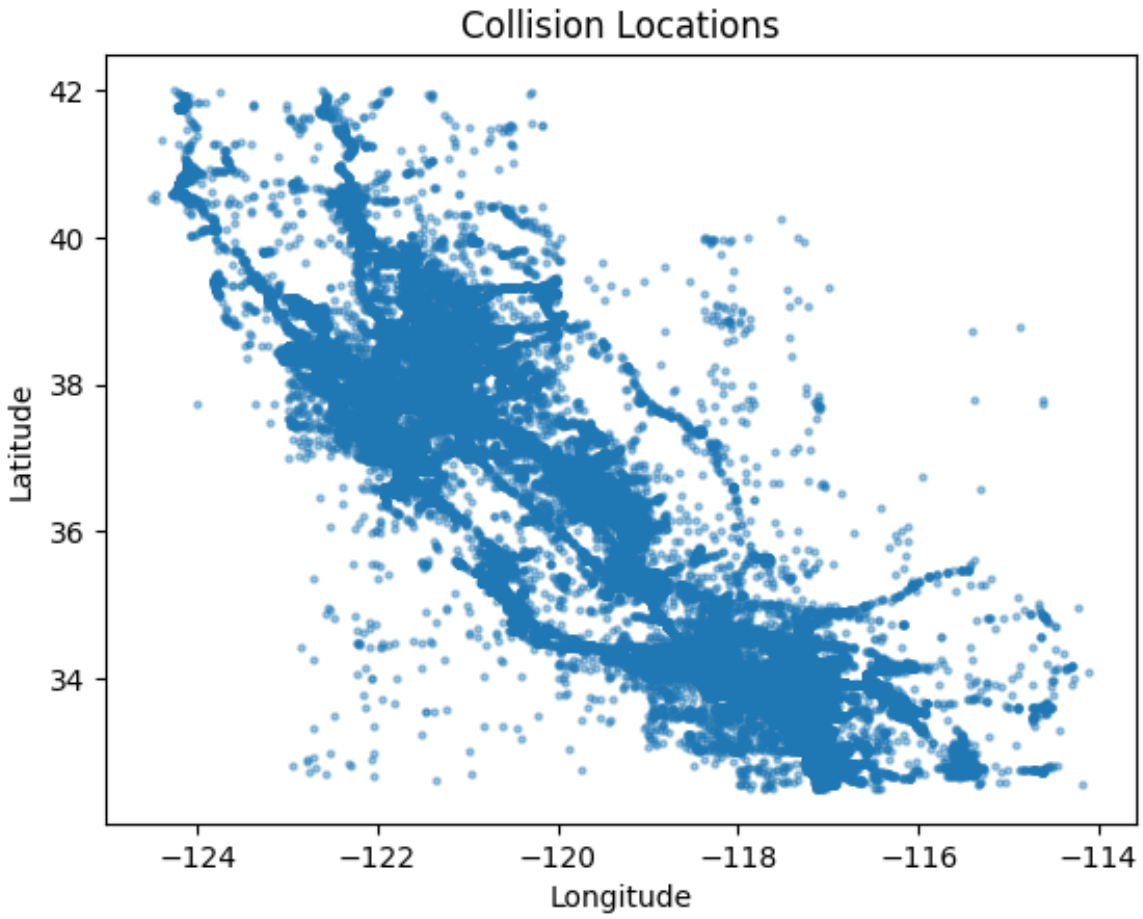
San Diego: 41,756

Alameda: 39,057



6**. Collision locations**

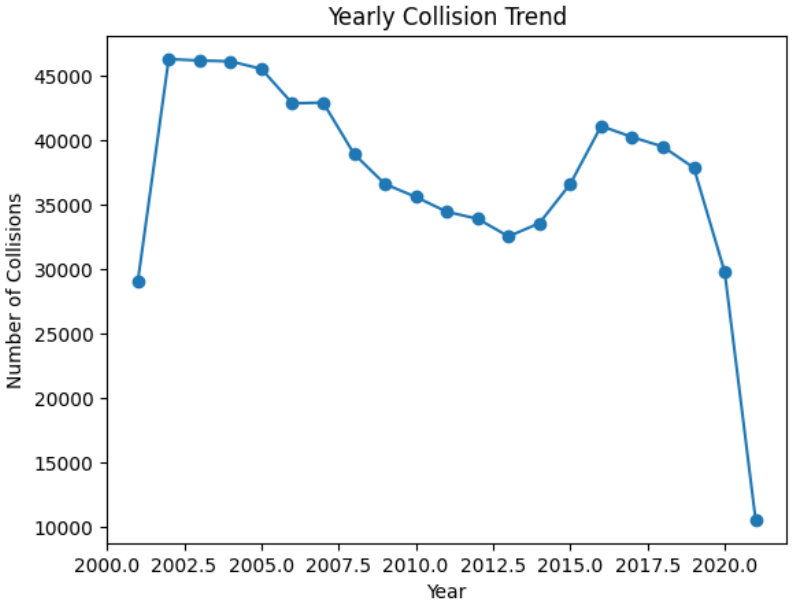
The scatter plot shows collision locations across California. Dense clusters appear around major population centers and transport corridors.



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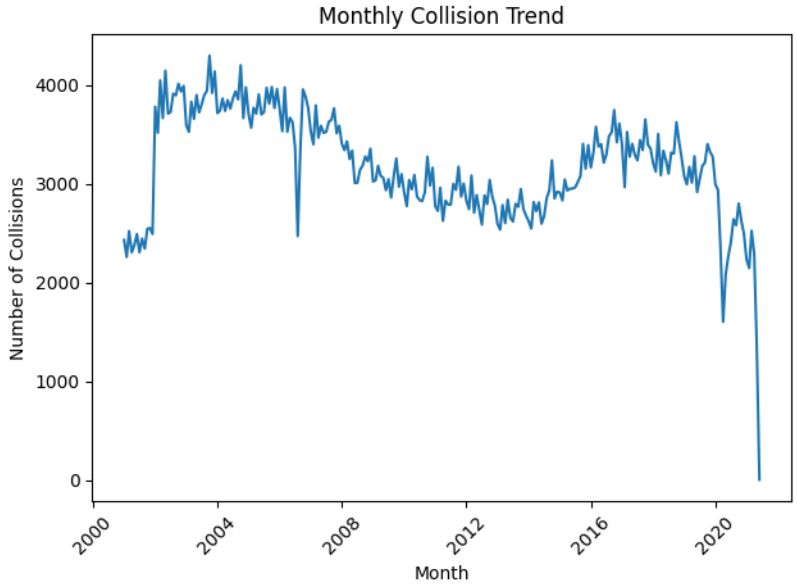
7. **Collision trends over time**

Yearly Collision Trends: The yearly chart is highest in the early 2000s, declines over much of the following decade, rises around 2015 to 2016, and then falls in the last years shown.



Recorded collision counts by year.

Monthly Collision Trends: The monthly series changes from year to year. The separate month-of-year query identifies October as the highest month, with 69,599 collisions.



Monthly collision counts across the available years.

8. Query results and next steps

Hourly result seems wrong

The saved query assigns all 780,981 records to 00:00.

4.1: **Top counties:**

Los Angeles 253,556;

Orange 62,734;

San Bernardino 44,830;

San Diego 41,756;

Alameda 39,057

4.2 Month with highest collisions: October - 69,599

4.3 Weather condition with highest collision: Clear - 647,360

4.4 Fatal collision share: 0.0%

4.5 Dangerous time for collision: 00:00

Final insights

1. Los angeles has highest number of collisions

2. Collision peaked in October month
3. In clear weather conditions, collisions are most frequent
4. Most collisions occured in daylight.
5. Most collisions happened in 0th hour.

#Recommendations

1. Review collision hotspots in Los Angeles County and compare them with traffic volume before setting priorities.
2. Check why on clear day most accidents are happening. Do we need to impose speed limits.
3. Investigate why most collisions happen during the 0th hour and consider implementing measures to improve safety during this time.
4. Analyze the reasons behind the peak in collisions during October and take preventive actions accordingly.
5. Investigate why most collisions occur in daylight and consider measures to enhance safety during these hours.